

# What linear programming contributes: World Food Programme experience with the “Cost of the Diet” tool

Romeo Frega, Jose Guerra Lanfranco, Sam De Greve, Sara Bernardini, Perrine Geniez, Nils Grede, Martin Bloem, and Saskia de Pee

## Abstract

**Background.** Linear programming has been used for analyzing children's complementary feeding diets, for optimizing nutrient adequacy of dietary recommendations for a population, and for estimating the economic value of fortified foods.

**Objective.** To describe and apply a linear programming tool (“Cost of the Diet”) with data from Mozambique to determine what could be cost-effective fortification strategies.

**Methods.** Based on locally assessed average household dietary needs, seasonal market prices of available food products, and food composition data, the tool estimates the lowest-cost diet that meets almost all nutrient needs. The results were compared with expenditure data from Mozambique to establish the affordability of this diet by quintiles of the population.

**Results.** Three different applications were illustrated: identifying likely “limiting nutrients,” comparing cost effectiveness of different fortification interventions at the household level, and assessing economic access to nutritious foods. The analysis identified iron, vitamin B<sub>2</sub>, and pantothenic acid as “limiting nutrients.” Under the Mozambique conditions, vegetable oil was estimated as a more cost-efficient vehicle for vitamin A fortification than sugar; maize flour may also be an effective vehicle to provide other constraining micronutrients. Multiple micronutrient fortification of maize flour could reduce the cost of the “lowest-cost nutritious diet” by 18%, but even this diet can be afforded by only 20% of the Mozambican population.

**Conclusions.** Within the context of fortification,

linear programming can be a useful tool for identifying likely nutrient inadequacies, for comparing fortification options in terms of cost effectiveness, and for illustrating the potential benefit of fortification for improving household access to a nutritious diet.

**Key words:** Economic access to nutritious food, fortification, linear programming, micronutrient inadequacy, Mozambique

## Introduction

Traditional methods that provide data to inform decisions on dietary recommendations usually have relied either on food intake recalls or on Food Balance Sheet (FBS) data. A mathematical approach known as linear programming offers a different, possibly more precise, approach by building on local food prices and food availability data to determine what theoretical diet can fulfill a population's nutrient needs at the lowest cost.

Linear programming has already been used in human nutrition to analyze and optimize children's diets during the complementary feeding period [1] or to optimize nutrient density in local diets [2, 3]. More recently, linear programming has been complemented by goal programming approaches to identify population-specific complementary feeding recommendations [4]. The approach has also previously been applied to analyze local food prices and to estimate the economic value of fortified food supplements [5].

As field practitioners, both Save the Children UK and the United Nations World Food Programme have applied this method over the past years to examine the role of economic constraints as one of the key underlying causes of malnutrition, and for WFP's programming. One of the applications has been to further inform feasibility studies of fortification. This article will give an overview of how linear programming can be useful for this purpose, with Mozambique as a practical example.

Romeo Frega, Jose Guerra Lanfranco, Sam De Greve, Sara Bernardini, Perrine Geniez, Nils Grede, Martin Bloem, and Saskia de Pee are affiliated with the United Nations World Food Programme, Rome; Martin Bloem and Saskia de Pee are also affiliated with the Friedman School of Nutrition, Tufts University, Boston, Massachusetts, USA.

Please direct queries to the corresponding author: Romeo Frega, Email: Romeo.Frega@gmail.com.

## Description of the method

### *How can linear programming be applied in the context of human nutrition?*

A user-friendly, Excel-based software for practitioners by Save the Children UK, linear programming has been used in a number of developing countries with the purpose of assessing to what extent economic constraints may be part of the nutrition problems. Save the Children UK and the United Nations World Food Programme have collected price data from different countries, such as Zambia, Mozambique, Djibouti, Bangladesh, Tanzania, and Niger, for this purpose.

The tool applied in this analysis ("The Cost of the Diet" or "CoD") is based on the World Food Composition Database, World Health Organization (WHO)-recommended nutrient requirements for different nutrients and for different age, sex, and physiological condition groups, and local food prices per 100 g, as well as a typical household member composition. In order to ensure that the diets that the tool calculates are acceptable or palatable in terms of quantities of the different foods, the tool offers the option to further customize the diet with constraints on the number of times a specific food can be included in the diet per week as well as on the portion size that is to be applied by the software every time the food is included. This works through two types of constraint: the "maximum constraints" represent the number of times a food can be included in the diet at most, and the "minimum constraints" represent the essential number of times a food must be included in the diet [1, 6]. Based on these factors, the software calculates a Minimum Cost Nutritious Diet, either for an entire household or for an individual child, by selecting foods from the list of available foods and optimizing their quantities at the

lowest possible cost. Here, it is important to point out that the resulting diet does not represent an actual diet, and it may not reflect the cultural nutritional habits of the local population. Yet it fulfills all the nutritional requirements (of a child or a household), it is optimized from a cost perspective, it is based on locally available foods, and it is theoretically palatable in terms of portion sizes and the number of servings per week; any other diet is either more expensive or not nutritionally adequate. **Figure 1** describes the inputs, calculation, and outputs of the method.

### **How linear programming can be used in discussions around fortification**

#### *Identification of limiting nutrients based on cost constraints*

The first application of linear programming within fortification feasibility studies is as a tool to identify possible "limiting nutrients," as identified by Briend and Ferguson [5]. **Table 1** shows the Minimum Cost Nutritious Diet for a household of five people in Gaza Province in Mozambique. In this case, the minimum constraint placed on each food was set to 0 and the maximum constraint to 21 portions per week (3 times per day times 7 days per week).

Looking at the diet in terms of percentage of nutrient requirements met can give a first indication of which nutrients are limiting the diet from a cost perspective. Energy and fat are included as equality constraints in the linear programming, meaning that the function will optimize by including these nutrients at 100% of the nutritional requirements. The rest of the nutrients are included as inequality constraints to be covered by at least 100% while minimizing the overall cost of the diet (also see Briend et al. [1] and Ferguson et al. [4]).

In the case of Gaza Province, the results show how

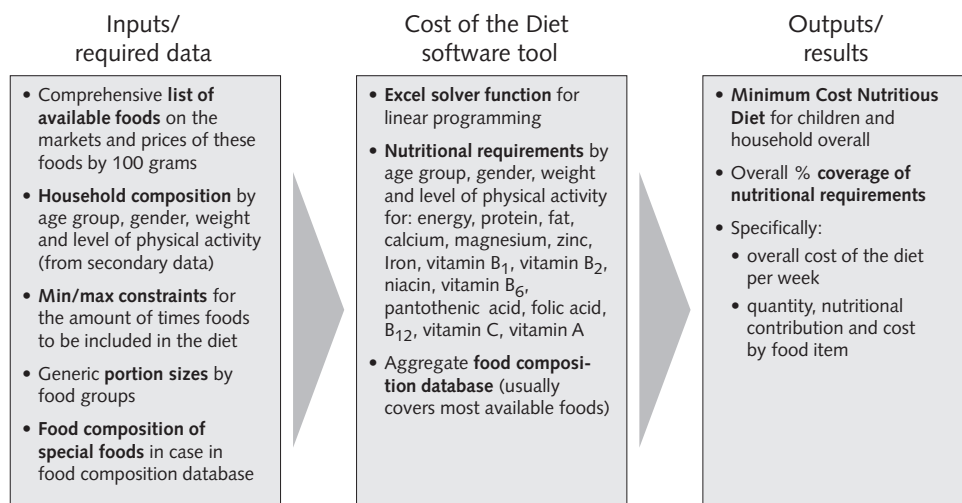


FIG. 1. Illustration of the cost of the diet calculation

TABLE 1. Example of a minimum-cost nutritious diet of a family in Gaza Province, Mozambique<sup>a</sup>

| Food list                                               | Quantity (g) | Cost (MZM) <sup>b</sup> | Energy (kcal)                   | Protein (g) | Fat (g) | Phytate (mg) | Vitamin A (RAE) | Vitamin C (mg) |
|---------------------------------------------------------|--------------|-------------------------|---------------------------------|-------------|---------|--------------|-----------------|----------------|
| Breastmilk                                              | 3,724        | 0.0                     | 2,421                           | 39.1        | 145.2   | 0.0          | 1,862.0         | 149.0          |
| Maize, dried, raw                                       | 307          | 5.6                     | 1,113                           | 24.9        | 11.1    | 2,459.6      | 0.0             | 0.0            |
| Cowpeas, whole dried                                    | 315          | 8.7                     | 964                             | 63.9        | 4.1     | 2,976.8      | 7.9             | 0.0            |
| Cowpeas, fresh                                          | 5,485        | 57.6                    | 16,785                          | 1,113.5     | 71.3    | 51,835.1     | 137.1           | 0.0            |
| Lemon                                                   | 105          | 0.8                     | 30                              | 1.2         | 0.3     | 0.0          | 1.6             | 55.7           |
| Coconut, ground                                         | 1,828        | 21.4                    | 3,876                           | 36.6        | 367.5   | 3,547.1      | 0.0             | 36.6           |
| Leaf, cowpea                                            | 1,047        | 7.7                     | 387                             | 38.7        | 2.1     | 439.6        | 2,716.0         | 345.4          |
| Leaf, amaranth, wild                                    | 315          | 3.8                     | 117                             | 11.7        | 0.6     | 132.3        | 817.4           | 104.0          |
| Beef, liver                                             | 262          | 23.6                    | 422                             | 63.9        | 12.8    | 0.0          | 53,328.9        | 39.3           |
| Fish, sardines, canned in oil                           | 59           | 10.2                    | 201                             | 14.5        | 15.7    | 0.0          | 39.6            | 0.0            |
| Sweet potato, white                                     | 525          | 6.4                     | 588                             | 12.6        | 0.5     | 304.5        | 0.0             | 73.5           |
| Yogurt                                                  | 261          | 25.3                    | 305                             | 18.2        | 16.9    | 0.0          | 127.7           | 7.8            |
| Vinegar                                                 | 105          | 3.2                     | 30                              | 1.2         | 0.3     | 0.0          | 1.6             | 55.7           |
| Cassava, dried, flour                                   | 2,320        | 48.9                    | 7,284                           | 60.3        | 16.2    | 3,015.5      | 162.4           | 1,670.1        |
| Cassava, raw                                            | 6,621        | 53.6                    | 8,673                           | 72.8        | 19.9    | 3,575.2      | 198.6           | 1,986.2        |
| Sugar, brown                                            | 1,828        | 49.2                    | 6,875                           | 0.0         | 0.0     | 0.0          | 0.0             | 0.0            |
| Vegetable oil                                           | 1,458        | 155.9                   | 12,571                          | 0.0         | 1,458.3 | 0.0          | 0.0             | 0.0            |
| Fish, small, dried, freshwater                          | 3,109        | 182.5                   | 10,414                          | 1,821.6     | 292.2   | 0.0          | 0.0             | 0.0            |
| Total                                                   | 29,674       | 665                     | 73,056                          | 3,395       | 2435    | 68,286       | 59,401          | 4,523          |
| Family composition (1 person in each category)          |              |                         | Weekly nutritional requirements |             |         |              |                 |                |
| Child (either sex) 12–23 mo                             |              |                         | 6,258.0                         | 76.3        | 208.6   | 0.0          | 2,800.0         | 210.0          |
| Child (either sex) 3–4 yr                               |              |                         | 8,400.0                         | 122.5       | 280.0   | 0.0          | 2,800.0         | 210.0          |
| Child (either sex) 7–8 yr                               |              |                         | 11,375.0                        | 189.0       | 379.2   | 0.0          | 3,500.0         | 245.0          |
| Man, 30–59 yr, 60 kg, vigorously active                 |              |                         | 24,150.0                        | 315.0       | 805.0   | 0.0          | 4,200.0         | 315.0          |
| Woman, 30–59 yr, 55 kg, vigorously active and lactating |              |                         | 22,872.5                        | 385.0       | 762.4   | 0.0          | 5,950.0         | 490.0          |
| Total weekly nutritional contribution                   |              |                         | 73,056                          | 1,088       | 2,435   | 0            | 19,250          | 1,470          |
| % requirements met                                      |              |                         | 100                             | 312         | 100     | 0            | 309             | 308            |

NE, Niacin Equivalents; RAE, Retinol Activity Equivalents

a. The diet is calculated separately for a child 12 to 23 months old as well as for a family of four. The results shown in this table are the aggregated results for a family of five, including the child of 12 to 23 months, since the cost for the entire family, including the infant, is relevant when it is compared with expenditure and income data. Notice the dependence of the diet on cowpeas (fresh) for the supply of niacin, folate, magnesium, iron, and zinc; and on liver for vitamin A, vitamin B<sub>2</sub>, and vitamin B<sub>12</sub>.

all other nutrients contained in the diet cover the daily requirements or provide more than that (i.e., nutrients covered more than 100% of the daily requirements). **The limiting nutrients in this example are iron and vitamin B<sub>2</sub>, as just around 100% of the nutritional requirements for these nutrients are met.** This means that given the current market prices of the available foods as well as their nutrient content, the solver function had to continue to add foods and increase the cost of the diet in order to meet the requirements of iron and vitamin B<sub>2</sub>; i.e., these two nutrients are “limiting” the diet and are therefore also the most expensive ones to access for people in this setting.

The example of Gaza shows us how certain nutrients limit the cost of the diet; however, overall 100% of all nutritional requirements of a household can theoretically be fulfilled—assuming that households have

sufficient income to afford this combination of foods and also assuming that households would consume the diet as indicated (including the frequent consumption of the highly nutritious components, such as beef liver). Another possible outcome of the linear optimization analysis could be that the availability of foods is so limited that the linear programming function will not be able to find a solution that reaches 100% or more on all nutrients. Such a scenario indicates a problem of food or nutrient availability on the markets that goes beyond problems of purely economic access, as people would not be able to access the necessary foods, regardless of their available income.

The identification of “limiting nutrients” using linear programming needs to be distinguished from an analysis where actual micronutrient inadequacies and deficiencies are identified based on food consumption

| Vitamin B <sub>1</sub> (mg) | Vitamin B <sub>2</sub> (mg) | Niacin (mg NE) | Vitamin B <sub>6</sub> (mg) | Folic acid (µg) | Vitamin B <sub>12</sub> (µg) | Pantothenic acid (mg) | Calcium (mg) <sup>c</sup> | Magnesium (mg) | Iron (mg) <sup>c</sup> | Zinc (mg) |
|-----------------------------|-----------------------------|----------------|-----------------------------|-----------------|------------------------------|-----------------------|---------------------------|----------------|------------------------|-----------|
| 0.8                         | 1.3                         | 16.1           | 0.3                         | 316.5           | 3.6                          | 6.7                   | 333.7                     | 130.3          | 0.2                    | 4.5       |
| 1.2                         | 0.6                         | 14.0           | 0.9                         | 76.9            | 0.0                          | 1.3                   | 4.6                       | 390.5          | 0.6                    | 5.5       |
| 1.7                         | 0.5                         | 17.3           | 0.8                         | 1,729.4         | 0.0                          | 3.4                   | 49.6                      | 441.0          | 1.2                    | 10.7      |
| 29.1                        | 8.8                         | 300.8          | 14.3                        | 30,113.7        | 0.0                          | 59.2                  | 863.9                     | 7,679.3        | 21.7                   | 186.5     |
| 0.0                         | 0.0                         | 0.3            | 0.1                         | 11.6            | 0.0                          | 0.2                   | 12.3                      | 8.4            | 0.0                    | 0.1       |
| 0.7                         | 0.2                         | 12.5           | 0.5                         | 292.5           | 0.0                          | 3.3                   | 65.8                      | 347.4          | 1.5                    | 12.8      |
| 0.9                         | 2.0                         | 13.6           | 5.7                         | 1,088.5         | 0.0                          | 2.6                   | 110.4                     | 648.9          | 1.9                    | 4.2       |
| 0.3                         | 0.6                         | 4.1            | 1.7                         | 327.6           | 0.0                          | 0.8                   | 33.2                      | 195.3          | 0.6                    | 1.3       |
| 0.5                         | 9.4                         | 42.3           | 1.4                         | 759.7           | 288.2                        | 14.9                  | 5.9                       | 52.4           | 2.0                    | 16.0      |
| 0.0                         | 0.2                         | 7.6            | 0.3                         | 4.7             | 5.3                          | 0.3                   | 104.0                     | 30.7           | 0.2                    | 1.8       |
| 0.6                         | 0.2                         | 4.9            | 1.4                         | 94.5            | 0.0                          | 1.9                   | 11.8                      | 157.5          | 0.3                    | 3.2       |
| 0.2                         | 0.8                         | 3.4            | 0.2                         | 36.5            | 2.1                          | 2.1                   | 223.6                     | 67.8           | 0.1                    | 3.1       |
| 0.0                         | 0.0                         | 0.3            | 0.1                         | 11.6            | 0.0                          | 0.2                   | 8.7                       | 8.4            | 0.0                    | 0.1       |
| 7.2                         | 1.2                         | 46.4           | 16.2                        | 835.1           | 0.0                          | 16.2                  | 266.8                     | 556.7          | 2.6                    | 16.2      |
| 8.6                         | 1.3                         | 56.3           | 19.2                        | 993.1           | 0.0                          | 19.2                  | 314.5                     | 662.1          | 3.2                    | 19.9      |
| 0.2                         | 0.2                         | 1.8            | 0.5                         | 18.3            | 0.0                          | 2.0                   | 497.3                     | 530.2          | 2.1                    | 3.7       |
| 0.0                         | 0.0                         | 0.0            | 0.0                         | 0.0             | 0.0                          | 0.0                   | 0.0                       | 0.0            | 0.0                    | 0.0       |
| 3.1                         | 8.4                         | 594.8          | 12.7                        | 870.4           | 373.0                        | 47.9                  | 16,910.8                  | 4,352.0        | 8.5                    | 161.6     |
| 55                          | 36                          | 1136           | 76                          | 37,580          | 672                          | 182                   | 19,817                    | 16,259         | 47                     | 451       |

|      |      |       |      |         |      |      |         |         |      |      |
|------|------|-------|------|---------|------|------|---------|---------|------|------|
| 3.5  | 3.5  | 42.0  | 3.5  | 1,050.0 | 6.3  | 14.0 | 1,050.0 | 420.0   | 4.1  | 28.7 |
| 3.5  | 3.5  | 42.0  | 3.5  | 1,050.0 | 6.3  | 14.0 | 1,050.0 | 420.0   | 4.1  | 28.7 |
| 6.3  | 6.3  | 84.0  | 7.0  | 2,100.0 | 12.6 | 28.0 | 1,470.0 | 700.0   | 6.2  | 39.2 |
| 8.4  | 9.1  | 112.0 | 9.1  | 2,800.0 | 16.8 | 35.0 | 2,100.0 | 1,820.0 | 9.6  | 49.0 |
| 10.5 | 11.2 | 119.0 | 11.2 | 3,500.0 | 19.6 | 49.0 | 2,100.0 | 1,890.0 | 23.2 | 59.5 |

|     |     |     |     |        |      |     |       |       |     |     |
|-----|-----|-----|-----|--------|------|-----|-------|-------|-----|-----|
| 32  | 34  | 399 | 34  | 10,500 | 62   | 140 | 7,770 | 5,250 | 47  | 205 |
| 171 | 106 | 285 | 223 | 358    | 1091 | 130 | 255   | 310   | 100 | 220 |

b. 1 MZM (Mozambican Metallic)= US\$0.038.

c. Refers to the amount absorbed (estimations based on WHO/FAO [7]). For calcium, we have used 0.25 for legumes, roots, tubers, and grains; 0.05 for high-oxalate foods (e.g., spinach); 0.45 for other fruits and vegetables; 0.32 for all other foods (including dairy products); and 0.3 for calcium supplements. For iron, we have used 0.5 for breastmilk; 0.2 for red meat; 0.1 for white flesh (chicken or fish); 0.05 for micronutrient powders (ferrous fumarate, possibly coated, or ferrous fumarate + NaFeEDTA); and 0.03 for plant-source foods and fortified foods.

surveys and biochemical indicators, respectively. Linear programming itself does not use any empirical household food consumption data. However, when a comparison with income or expenditure data shows that households are not actually able to afford the price of a Minimum Cost Nutritious Diet, theoretically these households will not have sufficient access to these “limiting nutrients,” as any diet different from the Minimum Cost Nutritious Diet either would be more expensive, and hence even less accessible, or would not fulfill the nutritional requirements, especially for the nutrients that are “limiting” the diet. Whether the “limiting nutrients” identified by linear programming actually correlate with the nutrient gap identified by food consumption surveys or biochemical indicators remains to be verified.

#### **Comparison of cost-effectiveness of different fortification vehicles**

As a second application in the fortification context, similar to the approach presented by Briend and Ferguson based on children aged 3 to 6 years [5], linear programming can also be used to show the economic value of fortification interventions to households. Once the Minimum Cost Nutritious Diet has been calculated and costed by the tool, it is possible to model the impact of the addition of a fortified food. If a fortified product is added to the linear programming tool, then the tool is likely to include that food in the cost-optimized diet, as this food will provide a higher amount of nutrients, especially micronutrients, per unit cost. In other words, it is a cost-effective food. The inclusion of these fortified foods will allow the tool to replace some of the more expensive sources of these nutrients, and hence

the inclusion of these products lowers the Minimum Cost of a Nutritious Diet.

Depending on the nutritional specifications of the fortified product, the cost impact on the Minimum Cost Nutritious Diet can be significant, especially if looked at from a household budget perspective. In the example of Mozambique, the cost-reduction impact of wheat flour fortification is compared with that of maize flour fortification, and the fortification of edible oil is compared with that of sugar (see **table 2**).

The example shows how fortification with vitamin A can lower the overall Minimum Cost of a Nutritious Diet by 1% if vegetable oil is fortified, whereas the fortification of sugar does not have an impact on the cost under the Mozambique conditions. The reason why sugar is not selected by the tool, even though it is fortified with vitamin A, is that the tool has no difficulty in filling the energy requirement of the family, as this requirement is already covered through the other components. Hence the sugar does not add value to the diet beyond the vitamin A that is added through the fortification of the sugar. But vitamin A can be added to the diet more cheaply than by adding it to sugar, given the price of sugar in Mozambique, for example, by adding beef liver or amaranth leaves, components that are rich in vitamin A and also add nutrition beyond vitamin A. The fortified vegetable oil, on the other hand, also helps to fulfill the fat requirement, and the vitamin A is an additional, “free” addition to the fulfillment of the nutritional requirements—hence it can replace some of the vitamin A, which would have to come at a price through other components of the diet. More notably, in the case of flour fortification with a multimicronutrient premix, fortified wheat flour lowers the price of the Minimum Cost Nutritious Diet by 12%, whereas fortified maize flour lowers the family food expenses by up to 18%. This significant cost reduction is due to the fact that micronutrients like vitamin B<sub>12</sub> or iron, which are added to the diet in a cheap way through fortification, are usually only available in more costly foods—in the

case of the example from Mozambique, in beef liver and small dried fish. Once the fortified food is included in the diet, the quantity of the more costly items can be reduced significantly. Obviously, this statement is true if those two flours can be easily and efficiently fortified and made accessible to the whole population.

These examples show how linear programming can be used to add the household budget perspective to a fortification feasibility study. It can make a strong case for fortification from a household perspective, as it exemplifies the possible cost reduction of a diet that meets household members’ nutrient requirements, and it can also be useful to compare different vehicles for fortification from an economic effectiveness point of view.

An alternative application of the methodology in this respect is the comparison of the cost effectiveness of different fortification formulations. Depending on the micronutrient composition of fortified foods included in the food list of the tool, the tool will indicate different levels of cost reductions for the household budget.

***Demonstrating how fortification can improve access to nutritious food at the household level***

As a third application in the context of fortification, the Minimum Cost of a Nutritious Diet can be used similarly to a poverty line. In order to estimate whether a population can have economic access to a nutritious diet, the cost of the Minimum Cost Nutritious Diet can be compared against income or expenditure data to estimate the minimum proportion of households who are not able to afford a nutritious diet. Ideally, household survey data are used for this purpose, but recent, good-quality, secondary household expenditure data can also be used.

The use of primary data is preferable for a variety of reasons. First, it can ensure that the methods by which the food expenditure data are collected correspond to the methods by which the data on cost of the diet are collected from the markets. It is important to make

TABLE 2. Fortification formulas applied in the case of Mozambique<sup>a</sup>

| Diet                              | Fortification formula                                                                            | Cost (MZM <sup>b</sup> /family/wk) | Cost reduction (%) |
|-----------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------|--------------------|
|                                   | No fortified product                                                                             | 664.5                              | N/A                |
| Including fortified vegetable oil | 1,000 µg RAE/100 g                                                                               | 658.0                              | 1                  |
| Including fortified sugar         | 1,000 µg RAE /100 g                                                                              | 664.5                              | 0                  |
| Including fortified wheat flour   | 1 µg/100 g vitamin B <sub>12</sub> , 4 mg/100 g iron, 2.2 mg/100 g zinc, 200 µg/100 g folic acid | 584.0                              | 12                 |
| Including fortified maize flour   | 1 µg/100 g vitamin B <sub>12</sub> , 4 mg/100 g iron, 2.2 mg/100 g zinc, 200 µg/100 g folic acid | 544.5                              | 18                 |

RAE, Retinol Activity Equivalents

a. For both vitamin A and iron, the absorption was adjusted to take into account lower absorption from fortified foods. For vitamin A, RAE is used; for iron, a 3% absorption factor is used to adjust the content in the fortified products [7].

b. MZM (Mozambican Metallic) = US\$0.038.



sure that all the foods that are self-produced are actually monetized as income or expenditure, respectively, as the Minimum Cost Nutritious Diet assumes that all foods are actually accessed through purchase on the markets, paying the market prices. Second, primary data allow a better estimate to be made of the proportions of households that can or cannot afford the nutritious diet. Finally, primary data can be collected specifically for a region overlapping geographically with the area for which the Minimum Cost of a Nutritious Diet has been calculated.

This analysis can also provide a strong argument for fortification. One can either use food expenditure data directly, or if these data are not available, one can use overall expenditure and make an assumption of what percentage of overall expenditure households could realistically be expected to spend on food (typically a maximum of 70%) and correct overall expenditure in that way. The challenge in doing this type of analysis is that income data are difficult to obtain and data collection is often not standardized; hence expenditure data are often used as a proxy for income.

**Figure 2** illustrates this analysis for the case of Mozambique. If we measure the food and beverage expenditure data according to national statistics against the Minimum Cost of a Nutritious Diet introduced in **table 1**, we can clearly see that for Mozambique, the data indicate that only the richest wealth group (quintile 5) can afford the Minimum Cost of a Nutritious Diet. The figure also illustrates how the Minimum Cost of a Nutritious Diet can be brought down by 18% if maize flour is fortified, bringing it within closer reach of the second wealth group (quintile 4). The analysis also shows how fortification alone cannot solve the problem of accessibility of a nutritious diet in Gaza, as the Minimum Cost Nutritious Diet (which is an optimistic estimation of the cost, as it assumes that people

are making the cost optimal choice for a nutritious diet), is still out of reach even if wheat flour is fortified. In order to further increase access to a nutritious diet, either the price of certain food items could be reduced subsidies, or other types of social safety nets could be established—such as the distribution of specific complementary food supplements or similar transfers.

## Conclusions

Linear programming, which has been used successfully for assessing the extent to which affordability issues affect access to a nutritious diet and for proposing cost-efficient diets that meet nutrient requirements, can also be a useful tool in fortification feasibility studies. First, it can help to identify which nutrients are likely to be “limiting nutrients” based on the composition and cost of locally available foods compared with nutrient needs. Second, it is useful for comparing the cost effectiveness of fortification interventions, as it demonstrates, for example, how cost-effective flour fortification can be in settings similar to those in Mozambique, by expressing the cost reduction of a Minimum Cost Nutritious Diet. Third, it can be used to estimate the proportion of households that are unlikely to be able to afford a diet that meets their nutrient requirements, for varying minimum cost, depending on whether fortified foods are included.

## Acknowledgments

Special thanks go to Esther Busquet, Rachel Evens, and Alex Rees at Save the Children UK for their continued effort and collaboration on this topic.

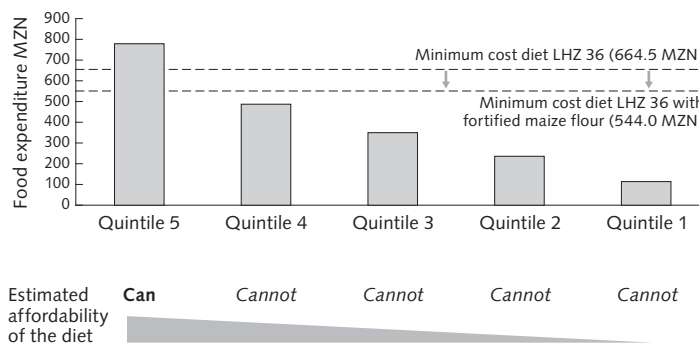


FIG. 2. Comparison of weekly expenditure on food products with Minimum Cost Nutritious Diets in Mozambique. The calculation is based on the monthly food and expenditure data from the Inquérito Sobre Orçamento Familiar, 2008/09, which were adjusted to a weekly level by dividing by 4.3 [8]; 1 MZM (Mozambican Metalic) = US\$0.038

## References

1. Briend, A, Darmon N, Ferguson E, Erhardt J. Linear programming: a mathematical tool for analyzing and optimizing children's diets during the complementary feeding period. *J Pediatr Gastroenterol Nutr* 2003; 36:12–22.
2. Darmon N, Ferguson E, Briend A. A cost constraint alone has adverse effects on food selection and nutrient density: an analysis of human diets by linear programming. *J Nutr* 2002;132:3764–71.
3. Darmon N, Ferguson E, Briend A. Linear and nonlinear programming to optimize the nutrient density of a population's diet: an example based on the diets of preschool children in Malawi. *Am J Clin Nutr* 2002;75:245–53.
4. Ferguson E, Darmon N, Fahmida U, Fitriyanti S, Harper T, Premachandra I. Design of optimal food-based complementary feeding recommendations and identification of key «problem nutrients» using goal programming. *J Nutr* 2006;136:2399–404.
5. Briend A, Ferguson E, Darmon N. Local food price analysis by linear programming: a new approach to assess the economic value of fortified food supplements. *Food Nutr Bull* 2001;22:184–9.
6. Save the Children UK (SCUK). Online Library. The cost of the diet—A practitioner's guide. Available at: [http://www.savethechildren.org.uk/en/54\\_9288.htm](http://www.savethechildren.org.uk/en/54_9288.htm). Accessed 7 May 2012.
7. World Health Organization/Food and Agriculture Organization. Guidelines on food fortification with micronutrients. 2006. Available at: [http://www.who.int/nutrition/publications/guide\\_food\\_fortification\\_micronutrients.pdf](http://www.who.int/nutrition/publications/guide_food_fortification_micronutrients.pdf). Accessed 7 May 2012.
8. Instituto Nacional de Estadísticas. Inquérito Sobre Orçamento Familiar, 2008/09. Quadros Básicos. Available at: <http://www.ine.gov.mz/ResourceCenter/Default.aspx>. Accessed 12 July 2012.